SRM Improves Layer-Wise LLM Encoding Across Cortical Networks During Naturalistic Story Comprehension

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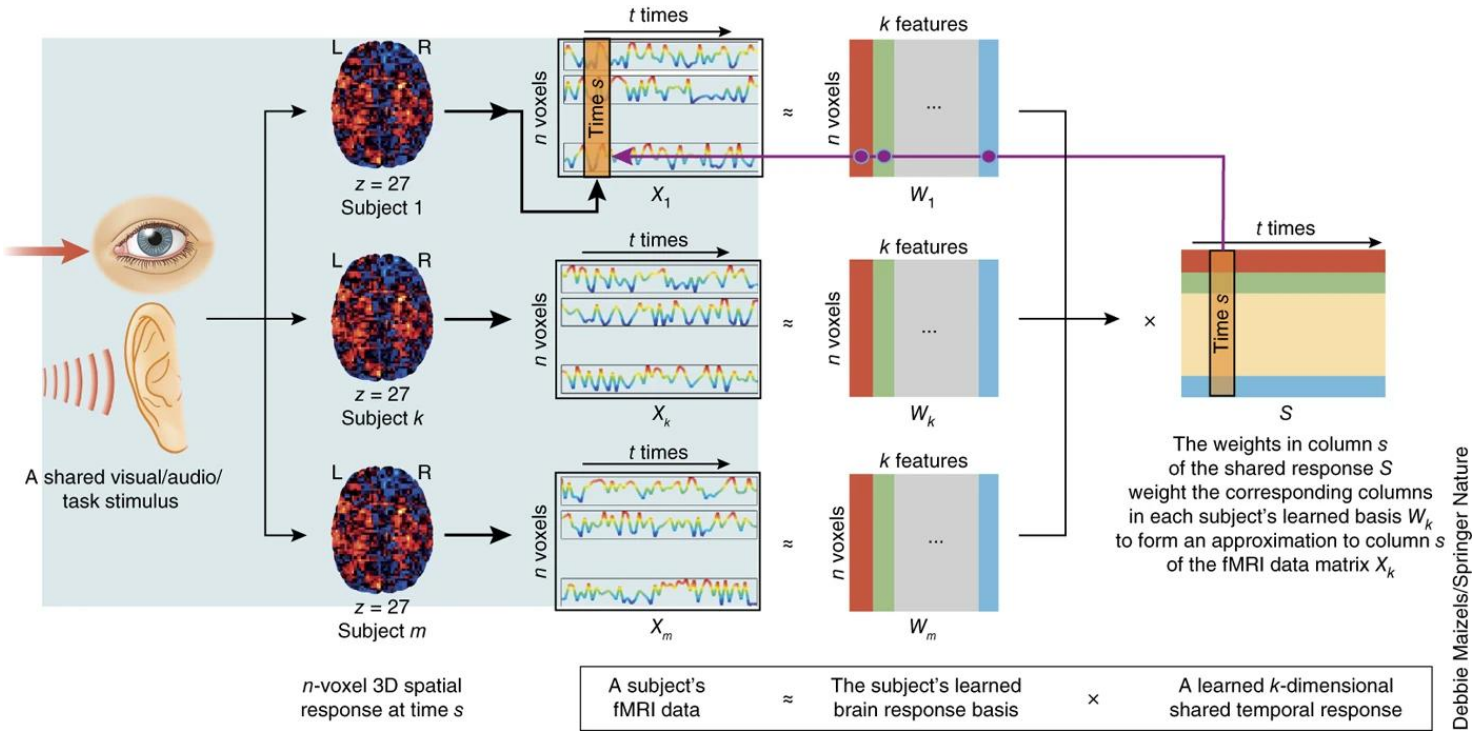
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Introduction

1. Recent studies suggest that large language model (LLM) representations can predict human brain responses during naturalistic language comprehension (Kumar et al., 2024) .

2. **However,** conventional fMRI encoding analyses rely **on anatomically aligned voxel responses**, potentially **ignoring shared response patterns that are distributed** differently across individuals.

3. **We hypothesize that shared response modelling will reduce the idiosyncratic variability of brain responses, thereby improving the alignment with LLM representations of linguistic information.**



(Cohen et al., 2017)

Questions / Aims

1. Does shared response modeling (SRM; Chen et al., 2015) improve LLM encoding of fMRI responses?

2. Are SRM-related gains organized by LLM layer depth and cortical network?

3. Do these gains exhibit hemispheric asymmetry?

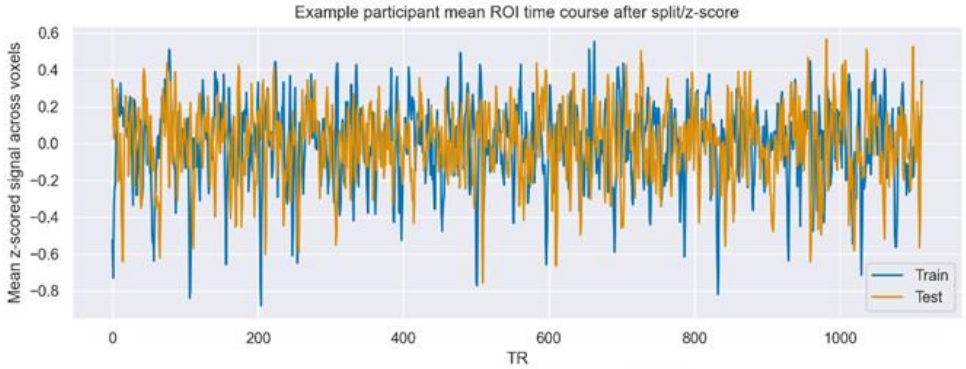
Methods

Encoding models were constructed for **(1) SRM-reconstructed** or **(2) raw BOLD signals** with LLM-derived embeddings as features.

25 participants from the "Narratives" **fMRI dataset** recorded during a passive story-listening task (Nastase et al., 2021).







Shared response modelling

$$X_i \approx W_i S$$

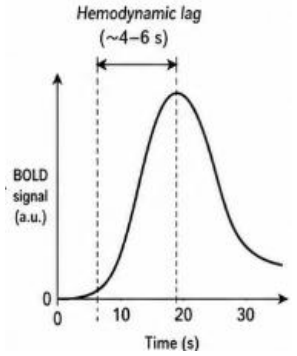
LLM embedding features

Encoding Model

(LeBel et al., 2023)

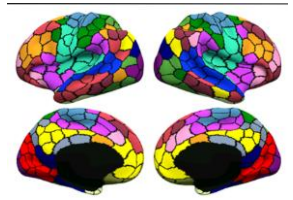


All layers (0-48) Contextual embeddings



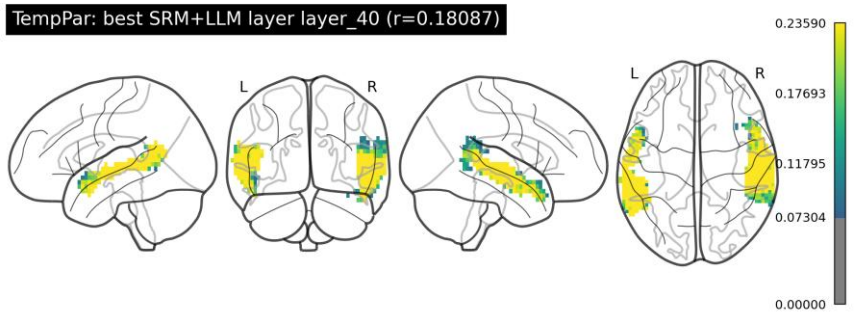
Fitted via ridge regression to predict:

- **raw anatomically aligned** fMRI responses **(RAW+LLM)**
- **SRM-reconstructed** fMRI responses **(SRM+LLM)**



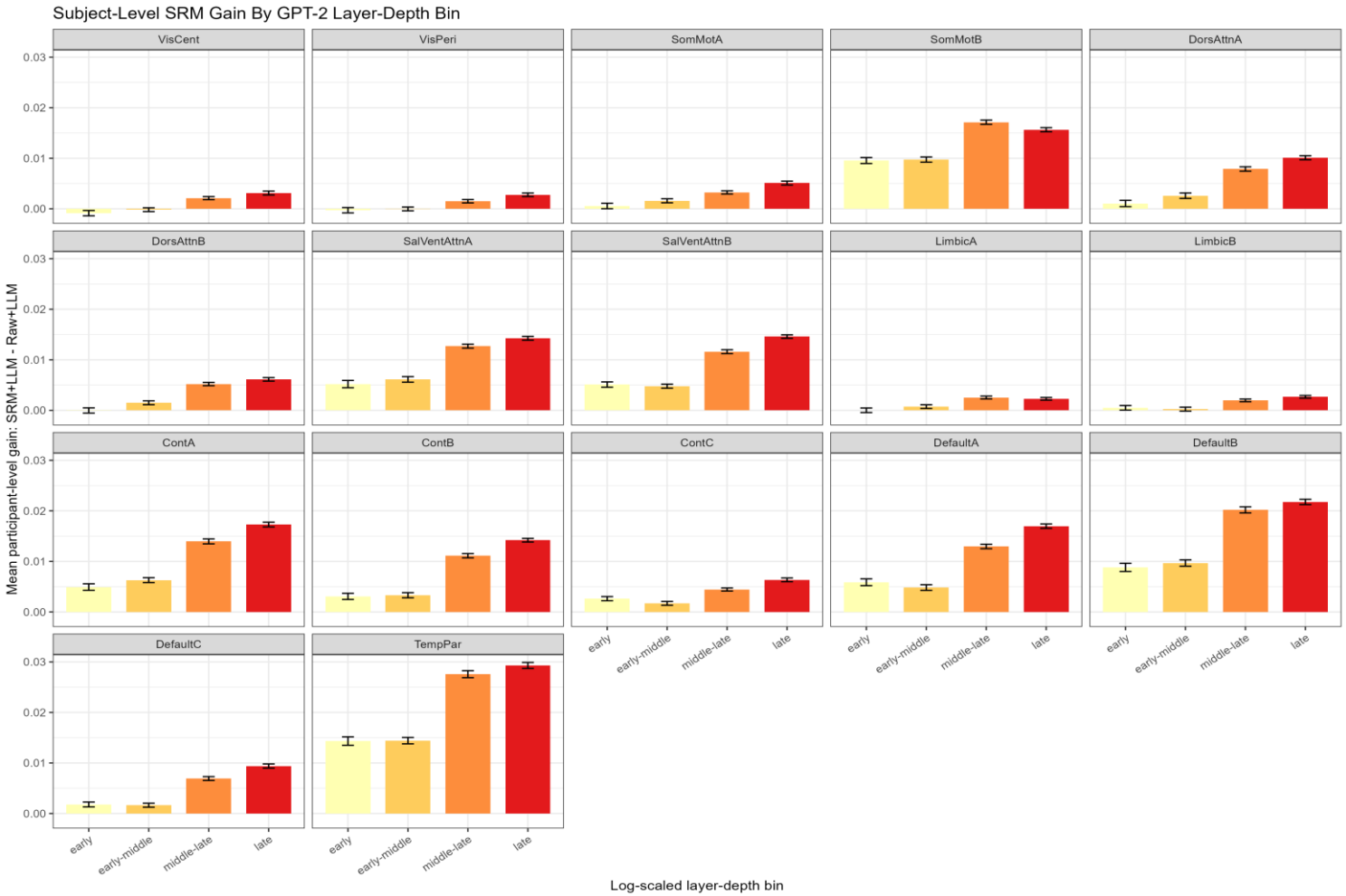
Schaefer-400 17 networks

Results



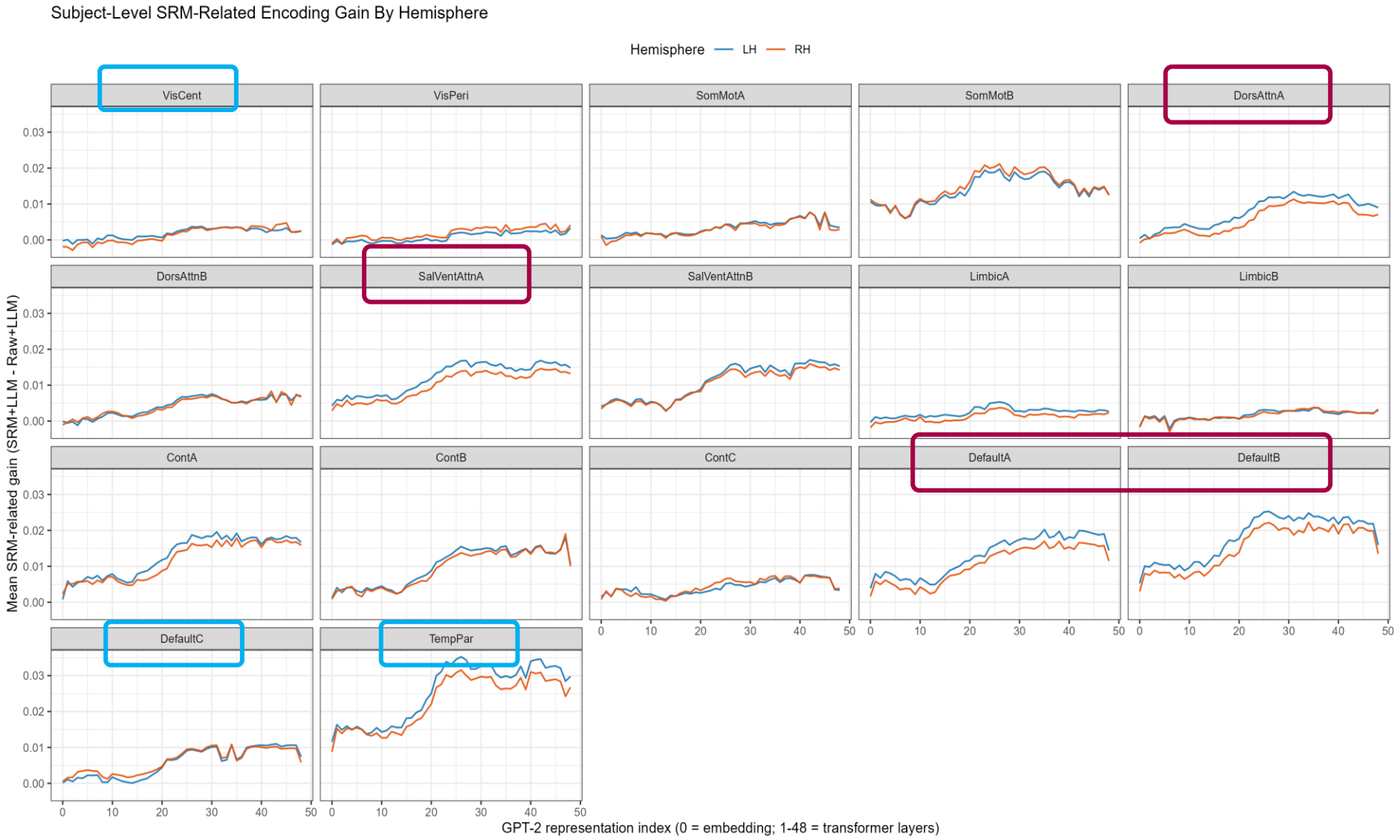
strongest SRM+LLM encoding in the temporo-parietal network at layer 40 (*r* = .18)

Effect	*F* statistic	FDR - *p*
layer depth	*F*(1, 24) = 160.40	< .00001
network	*F*(16, 384) = 40.54	< .00001
layer depth × network	***F*(16, 19184) = 251.24**	**< .00001**



At participant level, SRM-related gain showed a significant layer-depth-by-network interaction, *F*(16, 19,184) = 251.24, FDR *p* < .00001.Many networks showing stronger encoding in the middle-late or late depth.

Results



FDR-significant **LH > RH** SRM-related gain in participant-level layer bin

FDR-significant hemisphere × model layer interaction in SRM-related gain

Lateralization-index (LI):

Left-biased SRM-related gains across default-mode and attention/salience networks

LH > RH gains in DefaultA/B from **early-middle to late layers** and in DorsAttnA/SalVentAttnA at **middle-late and late layers**, while **VisCent, DefaultC, and TempPar** exhibited hemisphere-by-layer interactions

Conclusion

- **SRM+LLM** outperformed RAW+LLM, indicating that shared response modelling **reduces idiosyncratic neural response** patterns and enhances the predictability of neural dynamics from LLM representations.
- The SRM+LLM gain was strongest in the **temporo-parietal network** but also extended to **default-mode and attention/salience networks** in a layer and hemisphere-dependent manner.
- This provides a foundation for future work exploring how **functional alignment can clarify the relationship between LLM representations and shared neural responses** during naturalistic language comprehension.

References

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